
Trust Me or Check Me? Self-Reported Confidence, Verification, and Decision Authority in Language Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

1

2 1 Introduction

3 Large language models are increasingly asked not only to produce answers, but also to influence
4 whether those answers receive further scrutiny. A system may answer directly, seek additional
5 information, invoke a tool, defer to a human, or recommend that an answer be independently checked
6 before it is used. This creates a second-order decision problem beyond answer generation itself: *when*
7 *should an already-generated answer be used as is, and when should it be verified first?* When the
8 same AI system both generates an answer and participates in deciding whether that answer needs
9 checking, uncertainty becomes part of a decision-authority mechanism: the system can shape not
10 only what information is produced, but also when independent judgment enters the loop.

11 A natural control signal for this decision is the model’s own reported confidence. Suppose a model
12 has already produced an answer and reports a probability q that the answer is correct. If independent
13 verification costs C , while using a wrong answer without verification incurs cost L , then treating q as
14 the probability of correctness induces the simple expected-cost rule

$$(1 - q)L > C \implies \text{VERIFY_FIRST.}$$

15 This apparently straightforward rule hides two different questions. First, will a model’s downstream
16 verification behavior actually follow the policy implied by its stated confidence? Second, if it does,
17 will stronger agreement with that policy produce better decisions about which answers actually need
18 checking? These questions are distinct. A verification policy can be highly consistent with reported
19 confidence even when that confidence is poorly aligned with whether a particular answer is correct.

20 We study these questions with a controlled post-answer design that separates answer generation,
21 confidence elicitation, and verification policy. For each question, the model first produces a multiple-
22 choice answer, which is frozen. It then reports a probability q that this frozen answer is correct, which
23 is also frozen. Only afterward does the model choose between using the answer without verification
24 and verifying it first. At this final stage, we vary whether the previously reported q is hidden or
25 explicitly shown, whether verification authority is described as belonging to a human or to the AI
26 system, and the cost of using a wrong answer without verification. Because the answer and reported
27 confidence are held fixed before these manipulations, the experiment isolates changes in downstream
28 verification policy rather than changes in answer generation or confidence elicitation.

29 We evaluate 500 deterministic MMLU-Pro questions across four model families: OpenAI GPT-5.6
30 Sol, Anthropic Claude Sonnet 5, Google Gemini 3.8 Flash, and xAI Grok 4.20 non-reasoning.
31 For each model-question pair, the final decision crosses two decision-authority conditions, two
32 confidence-visibility conditions, and four error costs $L \in \{2, 5, 10, 20\}$, with verification cost fixed

33 at $C = 1$. This yields 32,000 primary verification decisions. Statistical comparisons use the question
34 as the independent sampling unit and question-level paired bootstrap confidence intervals.

35 Explicitly surfacing the model’s own previously reported confidence strongly changes verification
36 behavior, but not in a common direction across model families. For Claude Sonnet 5, visible
37 confidence increases verification by approximately 12.8–22.4 percentage points across authority and
38 cost conditions. For GPT-5.6 Sol, it decreases verification by 16.2–20.8 points throughout. Grok
39 4.20 similarly decreases verification by 8.6–16.4 points. Gemini 3.8 Flash shows smaller effects at
40 low error costs but substantially higher verification when confidence is visible at larger costs, with
41 increases of roughly 12–17 points at $L = 10$ and $L = 20$. Thus, the same intervention— making
42 self-reported confidence explicit to the downstream decision—can produce large and opposite policy
43 shifts across model families.

44 The most revealing case is GPT-5.6 Sol. We define *confidence-policy alignment* by comparing the
45 model’s final action with the action implied by its frozen reported confidence under the supplied cost
46 structure. When confidence is hidden, GPT-5.6 Sol disagrees with this confidence-implied policy
47 on roughly 19.6–27.2% of questions across authority and cost conditions. When the same reported
48 q is explicitly shown, disagreement collapses to 0.2–2.2%. Its behavior simultaneously becomes
49 almost perfectly ordered by the cost of error: stake-monotonicity violations fall from 95 to 1 per
50 1,000 trajectories.

51 Yet this increased alignment with reported confidence does not improve actual error filtering. At
52 the highest error cost, among GPT-5.6 Sol answers that are in fact wrong, the fraction used without
53 verification rises from 32.18% to 56.32% under human decision authority and from 33.33% to
54 54.02% under AI decision authority when confidence is made visible. In this case, explicitly surfacing
55 confidence therefore produces a policy that is substantially more consistent with the model’s stated
56 uncertainty while leaving substantially more of its high-cost errors unchecked. Greater confidence-
57 policy alignment and better verification quality are not the same property.

58 This distinction also matters when comparing models. A system can catch nearly all of its wrong
59 answers simply by verifying almost everything, but such a policy defeats the purpose of selective
60 verification. Conversely, reducing verification burden can leave more errors unchecked. We therefore
61 evaluate verification behavior using both error filtering and unnecessary verification of correct answers
62 rather than interpreting raw verification rate as a measure of quality.

63 Decision authority produces a smaller and less universal effect. Claude shows the clearest directional
64 shift, generally verifying more when the AI system rather than the human is described as controlling
65 the decision, particularly at higher error costs. The other model families show small or mixed average
66 effects. We therefore do not find a general human-versus-AI authority asymmetry across models.
67 Finally, on a prespecified 100-question robustness subset, the major confidence-visibility directions
68 persist under one semantically equivalent Stage-3 paraphrase, although we do not claim invariance to
69 arbitrary prompt wording.

70 These results concern model behavior rather than human responses. We do not measure whether
71 people trust, follow, resist, or override a model’s verification recommendation. Instead, we isolate
72 a technical prerequisite for such questions: whether a model’s own uncertainty can reliably govern
73 when its outputs are subjected to independent scrutiny. This matters for human–AI systems in which
74 an AI does more than provide information and also participates in determining when human judgment,
75 external evidence, or additional computation should intervene.

76 **Contributions.** Our contributions are threefold. First, we introduce a controlled post-answer
77 evaluation that freezes both answer and reported confidence before separately varying confidence
78 visibility, decision authority, and error cost across 32,000 matched verification decisions. Second,
79 we show that explicitly surfacing self-reported confidence can strongly reshape verification policy
80 in opposite directions across model families, cautioning against treating reported confidence as
81 a universally portable verification-control signal. Third, we identify a direct divergence between
82 confidence-policy alignment and verification quality: for GPT-5.6 Sol, visible confidence makes
83 downstream actions nearly coincide with the policy implied by reported q while substantially more
84 wrong high-cost answers remain unverified.

85 2 Related Work

86 **Self-reported uncertainty in language models.** A growing literature studies whether language
87 models can express useful uncertainty about their own answers. Early work showed that models
88 can be trained to verbalize probabilities that are meaningfully calibrated [Lin et al., 2022], while
89 subsequent studies examined black-box confidence elicitation in instruction-tuned and closed-source
90 models [Tian et al., 2023, Xiong et al., 2024]. These results motivate self-reported confidence
91 as an accessible uncertainty signal, particularly when token probabilities or model internals are
92 unavailable. At the same time, calibration is not sufficient for our setting: even a confidence score
93 that is informative about correctness may or may not govern a model’s later decision about whether
94 its answer should be checked.

95 **Confidence, abstention, and downstream action.** Recent work has begun to examine this gap
96 directly. RiskEval evaluates whether LLMs adapt abstention behavior as the penalty for an incorrect
97 answer changes, finding that models can fail to convert verbal uncertainty into appropriately risk-
98 sensitive abstention policies [Wang et al., 2026]. Kumaran [Kumaran, 2026] uses a two-stage answer-
99 confidence-abstention paradigm and finds that verbal confidence predicts subsequent commitment
100 more strongly than objective correctness, distinguishing verbal confidence from token-probability-
101 based uncertainty. More broadly, selective prediction and abstention methods use uncertainty to
102 trade coverage for error reduction, including approaches that explicitly control the error rate among
103 accepted answers [Dong and Shinnou, 2026].

104 Our experiment is complementary to this work. Rather than asking only whether reported confidence
105 predicts a later decision, we freeze the answer and reported probability and intervene on whether
106 that same probability is explicitly available at decision time. This permits a matched comparison of
107 the downstream policy with confidence hidden versus visible. It also separates *confidence-policy*
108 *alignment*—whether the action agrees with the decision rule implied by reported q —from *verification*
109 *quality*—whether wrong answers are filtered while unnecessary verification of correct answers is
110 avoided.

111 **Reliance and decision authority in human–AI systems.** Research on AI-assisted decision making
112 distinguishes mere reliance from *appropriate* reliance: accepting useful AI advice while rejecting
113 or scrutinizing erroneous advice [Schemmer et al., 2022]. Interface interventions can alter this
114 behavior; for example, cognitive forcing mechanisms can reduce human overreliance on incorrect AI
115 recommendations [Bućinca et al., 2021]. Our study does not measure human reliance. Instead, it
116 examines an upstream system-design question relevant to such interactions: when an AI participates
117 in deciding whether its own output deserves independent scrutiny, how do its reported uncertainty
118 and the location of decision authority shape that recommendation?

119 3 Experimental Setup

120 3.1 Dataset and models

121 We use the MMLU-Pro test split [Wang et al., 2024], a multiple-choice benchmark spanning 14
122 subject categories. The primary V2-B sample contains 500 questions: the earlier 200-question V2-A
123 sample plus 300 additional questions selected by deterministic category-stratified sampling without
124 reference to observed model results. The 500 questions constitute a single primary sample; V2-A and
125 V2-B are therefore not independent replications. A separate 100-question category-stratified subset,
126 also selected without reference to outcomes, is used for prompt robustness.

127 We evaluate four independently developed model families: OpenAI GPT-5.6 Sol (gpt-5.6-sol),
128 Anthropic Claude Sonnet 5 (claude-sonnet-5), Google Gemini 3.8 Flash (gemini-3.8-flash),
129 and xAI Grok 4.20 non-reasoning (grok-4.20-0309-non-reasoning). OpenAI used reasoning
130 effort none, Anthropic thinking was disabled, Gemini used thinking level low, and the xAI model
131 is explicitly non-reasoning. No tools, search, retrieval, code execution, or external context were
132 available. We used provider-native structured JSON outputs and did not override temperature or top- p
133 sampling parameters. Output was capped at 64 tokens for OpenAI, Anthropic, and xAI. Gemini used
134 a 512-token ceiling because its mandatory low-thinking tokens share the output budget.

3.2 Three-stage protocol

Each model–question pair proceeds through three stages. In Stage 1, the model receives the question and answer choices and returns exactly one multiple-choice answer. This answer is then frozen. In Stage 2, the model receives the same question, choices, and its frozen answer, is explicitly told not to change or re-solve the answer, and reports a probability $q \in [0, 1]$ that the frozen answer is correct. This reported confidence is also frozen.

Stage 3 does not permit the answer to change. Instead, the model selects exactly one of two actions: `USE_UNVERIFIED` or `VERIFY_FIRST`. Verification has fixed cost $C = 1$ and is defined to return the correct answer. Using the frozen answer without verification costs zero if it is correct and L if it is wrong, where

$$L \in \{2, 5, 10, 20\}.$$

These costs define a stylized decision problem rather than literal real-world harm.

For every frozen question, answer, and confidence value, Stage 3 crosses two decision-authority framings and two confidence-visibility conditions. Under *human authority*, the prompt states that the human controls whether the answer is independently verified before use; under *AI authority*, it states that the AI system controls verification before delivery to the human. The prompts use matched owner-specific wording and corresponding grammar. In the *visible-confidence* condition, Stage 3 additionally states the model’s previously reported probability q ; in the *hidden-confidence* condition, this sentence is omitted. Confidence is never re-elicited in Stage 3. Crossing two authority conditions, two visibility conditions, and four values of L yields 16 Stage 3 decisions per model–question pair, or 32,000 primary decisions in total.

3.3 Metrics

We report Stage 1 accuracy and the Brier score of Stage 2 confidence separately from Stage 3 verification behavior. For each model, authority condition, confidence condition, and error cost, we measure the verification rate, unverified use among actually wrong answers, and unnecessary verification among correct answers. The latter two distinguish error filtering from verification burden: a policy that verifies nearly everything can catch almost every error while providing little selectivity.

We also compare Stage 3 behavior with the action implied directly by reported confidence. Under the supplied cost model, the confidence-implied policy verifies when

$$(1 - q)L > C.$$

We call agreement between the model’s Stage 3 action and this action *confidence-policy alignment*, and report the corresponding disagreement rate. This is a behavioral comparison with the model’s *reported* probability; it does not assume that q represents a latent internal belief. We additionally measure stake-monotonicity violations, defined as `VERIFY_FIRST`→`USE_UNVERIFIED` reversals as L increases within an otherwise matched trajectory, as well as realized cost under the synthetic cost model.

The confidence-visibility effect is the paired difference

$$P(\text{VERIFY_FIRST} \mid \text{visible}) - P(\text{VERIFY_FIRST} \mid \text{hidden}),$$

and the owner effect is

$$P(\text{VERIFY_FIRST} \mid \text{AI authority}) - P(\text{VERIFY_FIRST} \mid \text{human authority}).$$

We report models individually rather than treating the 32,000 decisions as independent observations.

3.4 Statistical inference and prompt robustness

The independent sampling unit is the question. We use 5,000 question-level paired bootstrap resamples and report nominal 95% percentile confidence intervals, keeping all matched factor cells for a resampled question together. For metrics restricted to wrong answers, questions are resampled first and the wrong-answer subset is applied within each bootstrap replicate. We do not apply a multiplicity correction across the many descriptive cellwise comparisons; accordingly, we emphasize effect sizes, intervals, and repeated directional patterns rather than isolated significant cells.

179 Prompt robustness uses the fixed 100-question subset with the same frozen answers and reported
 180 confidences under one semantically equivalent Stage 3 paraphrase. The preregistered robustness
 181 analysis covered the hidden-confidence condition (3,200 decisions). After completion of the primary
 182 experiment, we ran the corresponding visible-confidence extension (3,200 additional decisions).
 183 We therefore distinguish the hidden-condition robustness analysis as preregistered and the visible-
 184 condition extension as post-primary.

185 References

- 186 Zana Bućinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. To trust or to think: Cognitive forcing
 187 functions can reduce overreliance on ai in ai-assisted decision-making. *Proceedings of the ACM*
 188 *on Human-Computer Interaction*, 5(CSCW1), 2021. doi: 10.1145/3449287.
- 189 Sijin Dong and Hiroyuki Shinnou. Uncertainty-aware abstention in large language models with
 190 provable alignment guarantees, 2026.
- 191 Dharshan Kumaran. Reported confidence in llms tracks commitment more than correctness, 2026.
- 192 Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in
 193 words. *Transactions on Machine Learning Research*, 2022.
- 194 Max Schemmer, Patrick Hemmer, Niklas Kühl, Carina Benz, and Gerhard Satzger. Should i follow
 195 ai-based advice? measuring appropriate reliance in human-ai decision-making, 2022.
- 196 Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea
 197 Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated
 198 confidence scores from language models fine-tuned with human feedback. In *Proceedings of the*
 199 *2023 Conference on Empirical Methods in Natural Language Processing*, pages 5433–5442, 2023.
 200 doi: 10.18653/v1/2023.emnlp-main.330.
- 201 Jiawei Wang, Yanfei Zhou, Siddhartha Devic, and Deqing Fu. Are llm decisions faithful to verbal
 202 confidence?, 2026.
- 203 Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming
 204 Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi
 205 Fan, Xiang Yue, and Wenhui Chen. MMLU-Pro: A more robust and challenging multi-task
 206 language understanding benchmark. In *Advances in Neural Information Processing Systems*,
 207 volume 37, 2024. doi: 10.52202/079017-3018.
- 208 Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can llms
 209 express their uncertainty? an empirical evaluation of confidence elicitation in llms. In *International*
 210 *Conference on Learning Representations*, 2024.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive” or “we were unable to find the license for the dataset we used”). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”.
- Keep the checklist subsection headings, questions/answers and guidelines below.
- Do not modify the questions and only use the provided macros for your answers.

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The authors should answer **[Yes]** if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.